

	TWIGS NATURALS	Version #:	1.0	
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Title:	TN-Laptop1 Security Camera Deployment, Cloudflare Tunnel Implementation , and Remote Monitoring Technical Implementation Guide	<u>AUTHOR</u> Jephete Fenelon	<i>Signature:</i> 	<i>Date:</i> 6/15/2026
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Total Pages:	47	<u>APPROVER:</u>	<i>Signature:</i>	<i>Date:</i>

Version Tracking Table:

Version	Date	Author	Department	Reviewer/ Approver	Summary of Changes	Status
1.0	6/15/2026	Jephete Fenelon	Information Technology	Nigel Jordan	Initial deployment, configuration, testing, and documentation of the TN-Laptop 1 security camera system.	Completed
1.1	6/19/2026	Jephete Fenelon	Information Technology	Nigel Jordan	Performed additional testing and validation of camera monitoring services. Documented external access investigation, local monitoring verification, and preliminary network troubleshooting activities. Updated project status to reflect continued external access evaluation and validation efforts.	Validation

1.2	6/20/2026	Jephete Fenelon	Information Technology	Nigel Jordan	.Added network configuration updates, reboot validation, external access testing, WAN/public IP analysis, router review, and updated investigation findings.	Operational
1.3	6/21/2026	Jephete Fenelon	Information Technology	Nigel Jordan	Added Cloudflare Tunnel evaluation, alternative remote access research, implementation planning, dependency identification, and updated project documentation and findings.	Research
1.4	6/22/2026	Jephete Fenelon	Information Technology	Nigel Jordan	Successfully completed Cloudflare Tunnel deployment, Cloudflare installation, tunnel validation, DNS research, published application route creation for cameras.twignaturals.com, CNAME endpoint identification, management updates, and project documentation.	Research & Validation
1.5	6/27/2026	Jephete Fenelon	Information Technology	Nigel Jordan	Completed Cloudflare Tunnel validation activities, installed and activated the Cloudflare connector on TN-Laptop1, verified tunnel connectivity and published application routing, and identified DNS configuration within Squarespace as the remaining requirement for external camera access.	Implementation & Validation
1.6	7/05/2026	Jephete Fenelon	Information Technology	Nigel Jordan	Documentation updated to include Cloudflare Tunnel troubleshooting activities, validation results, access dependency findings, and outstanding requirements for local camera service verification on TN-Laptop1.	Troubleshooting
1.7	7/07/2026	Jephete Fenelon	Information Technology	Nigel Jordan	Updated project status following a July 7, 2026 follow-up validation attempt. Documented continued Action1 Remote Connect permission limitations preventing further endpoint validation and confirmed the project remains in the Research and	Project Status Update

					Validation phase pending authorized endpoint access.	
1.8	7/08/2026	Jephete Fenelon	Information Technology	Nigel Jordan	Updated project documentation, documented the current project status, and prepared the technical handoff for final validation activities.	Technical Handoff
1.9	7/09/2026	Joshua Ugborugbo	Information Technology	Nigel Jordan	Completed endpoint validation using Action1 Remote Connect. Verified ContaCam configuration, camera operation, HTTP/HTTPS services, Cloudflare Tunnel service, Published Application, local monitoring functionality, and documented technical validation findings. Reviewed and signed implementation documentation.	Validation Completed
2.0	7/13/2026	Joshua Ugborugbo	Information Technology	Nigel Jordan	Completed DNS infrastructure investigation and root cause analysis. Validated authoritative DNS providers (Squarespace/NS1), confirmed inactive Cloudflare DNS zone due to invalid nameserver delegation, documented DNS findings, and updated project implementation status.	DNS Investigation
2.1	7/17/2026	Jephete Fenelon	Information Technology	Nigel Jordan	Final internship documentation revision completed. Updated the implementation guide to include the July 9 endpoint validation, July 13 DNS investigation, July 16 project continuation, updated project status, knowledge transfer documentation, project transition, and document maintenance to support ongoing implementation and future updates.	Implementation Guide Update

1. Document Ownership

This implementation guide was prepared and maintained by **Jephete Fenelon** throughout the **TN-Laptop1 Security Camera Deployment and Remote Monitoring Project** during my internship at **Twigs Naturals**.

The document records the planning, implementation, configuration, testing, troubleshooting, technical research, project communications, and documentation activities completed throughout the project. It serves as the official technical implementation guide and project record developed during my internship.

To provide a complete and accurate project history, this document also includes subsequent implementation updates following the documented technical handoff. These sections appropriately identify and attribute the contributions of team members who participated in the remaining validation, DNS investigation, and implementation activities after the project transitioned.

Unless otherwise noted, the implementation activities and technical documentation completed during my internship were prepared and documented by **Jephete Fenelon**. Subsequent updates are identified within the document and attributed to the appropriate contributing team member(s).

2. Purpose

The purpose of this document is to provide a comprehensive technical record of the planning, design, deployment, implementation, configuration, validation, troubleshooting, technical research, and operational assessment of the **TN-Laptop1 Security Camera Deployment and Remote Monitoring Project** utilizing **ContaCam**, **Cloudflare Zero Trust**, and **Cloudflare Tunnel** technologies.

This document serves as the official technical implementation guide and operational record for the project. It is intended to support system administration, operational maintenance, incident response, future enhancements, project continuity, and knowledge transfer. The documentation captures the complete implementation lifecycle, including system planning, endpoint preparation, software installation, security configuration, network validation, remote monitoring research, Cloudflare Tunnel deployment, DNS investigation, root cause analysis, technical findings, management communications, collaborative implementation activities, project status updates, and technical handoff.

Additionally, this document evaluates secure remote camera access solutions while preserving the organization's existing website, email, and DNS infrastructure. It provides future administrators with a complete implementation history, technical decisions, validation results,

troubleshooting activities, operational findings, recommendations, and the remaining infrastructure dependencies required to support successful production deployment and long-term operational maintenance.

This document also serves as a knowledge transfer resource by documenting the collaborative implementation efforts performed throughout the project, including the transition of the remaining validation and implementation activities following management direction, ensuring continuity for future project completion and ongoing support.

3. Project Scope

This project encompasses the complete planning, implementation, configuration, validation, troubleshooting, documentation, technical research, collaborative validation, and operational assessment of the **TN-Laptop1 Security Camera System** and its associated secure remote monitoring infrastructure.

The project includes the successful implementation of a browser-based local monitoring solution together with the research, deployment, validation, troubleshooting, and operational assessment of secure remote access technologies intended to support future production deployment while preserving the organization's existing network, website, email, and DNS architecture.

The scope also includes comprehensive endpoint validation, Cloudflare Zero Trust implementation, Cloudflare Tunnel deployment, DNS investigation, root cause analysis, technical documentation, project collaboration, management communications, technical handoff, and project continuity activities performed throughout the implementation lifecycle.

Included Within Scope

Project Planning and Implementation

- Project planning and solution design
- Technical implementation planning
- Endpoint preparation
- Project coordination
- Technical handoff and knowledge transfer

Endpoint Configuration

- ContaCam installation and configuration

- Camera deployment and operational validation
- Browser-based monitoring implementation
- Internal web server configuration
- Authentication and access control implementation
- Windows power management optimization
- Security hardening and operational controls

System Validation

- Functional testing and operational validation
- Local monitoring verification
- Camera performance validation
- Endpoint validation
- Browser accessibility testing
- HTTP and HTTPS service validation
- Authentication verification

Network and Infrastructure

- Windows Firewall configuration review
- Network configuration validation
- Router configuration review
- DHCP reservation validation
- Port forwarding configuration and testing
- WAN and public IP address analysis
- External connectivity investigation

Cloudflare Implementation

- Cloudflare Zero Trust deployment
- Cloudflare Tunnel deployment
- Cloudflare connector (**cloudflared**) installation
- Published Application route configuration
- Cloudflare Tunnel health validation
- Cloudflare dashboard validation
- Cloudflare service validation
- Tunnel connectivity verification

DNS Investigation

- DNS implementation research
- Authoritative DNS validation
- External DNS compatibility assessment
- DNS infrastructure investigation

- Root cause analysis
- Nameserver delegation assessment
- External hostname resolution validation

Documentation and Project Management

- Technical troubleshooting activities
 - Technical documentation development and maintenance
 - Project status reporting
 - Management communication
 - Operational recommendations
 - Implementation updates
 - Collaborative project support
 - Future implementation planning
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Excluded From Scope

The following activities are outside the scope of this implementation:

- Enterprise network redesign
 - ISP infrastructure modifications
 - Carrier network configuration
 - Router hardware replacement
 - VPN implementation
 - Long-term video storage expansion
 - Third-party monitoring platform integration
 - Procurement of a static public IP address
 - Migration of the organization's production DNS provider
 - Domain registrar migration
 - Production DNS modifications without management approval
 - Cloudflare production cutover without management authorization
 - Changes to the organization's existing website or email infrastructure
 - Administrative permission changes within the Action1 platform
 - Organization-wide infrastructure changes unrelated to the TN-Laptop1 Security Camera Deployment project
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4. Executive Summary

The **TN-Laptop1 Security Camera Deployment and Remote Monitoring Project** was initiated to design, implement, and document a secure, reliable, and continuously available

monitoring solution utilizing **ContaCam**, the integrated **HP TrueVision HD Camera**, **Cloudflare Zero Trust**, and **Cloudflare Tunnel** technologies. The primary objective of the project was to establish a browser-based monitoring platform capable of supporting continuous local monitoring while evaluating secure methods for future remote access without impacting the organization's existing website, email, or DNS infrastructure.

Implementation activities included endpoint preparation, software installation, camera configuration, authentication and security implementation, Windows power optimization, internal web server configuration, browser-based monitoring, operational validation, network assessment, technical troubleshooting, and comprehensive technical documentation. Throughout the implementation lifecycle, extensive testing was performed to verify camera functionality, recording services, browser accessibility, authentication controls, application stability, and continuous operation. These activities confirmed that the local monitoring solution was successfully implemented and fully operational.

To support secure remote monitoring requirements, additional implementation and research activities were conducted utilizing **Cloudflare Zero Trust** and **Cloudflare Tunnel** technologies. The **cloudflared** connector was successfully installed and registered with the Cloudflare Zero Trust environment. Tunnel connectivity was established and validated, tunnel health was confirmed, and a Published Application route for **cameras.twignaturals.com** was successfully configured. The implementation demonstrated that Cloudflare Tunnel provides a secure approach for remote access while preserving the organization's existing infrastructure.

Comprehensive investigation was performed to determine the cause of unsuccessful external camera access. Activities included Windows Firewall validation, router configuration review, DHCP reservation verification, port forwarding assessment, WAN connectivity analysis, public IP comparison, endpoint validation, Cloudflare Tunnel verification, DNS investigation, and multiple external connectivity tests. These investigations confirmed that the deployed monitoring solution, including the local camera services and Cloudflare Tunnel implementation, was functioning correctly within the local environment.

As the project progressed, additional DNS validation and root cause analysis determined that the remaining external accessibility limitation was not related to the endpoint configuration or Cloudflare Tunnel deployment. Instead, the investigation confirmed that the production domain remained under the authoritative DNS management of Squarespace/NS1, while the Cloudflare DNS zone was inactive due to invalid nameserver delegation. As a result, the hostname **cameras.twignaturals.com** could not be publicly resolved until the required DNS configuration is completed within the authoritative DNS provider.

During the implementation, additional endpoint validation required direct access to **TN-Laptop1** through the **Action1 Remote Connect** feature. Because the assigned Action1 account did not have Remote Connect permissions for the TN endpoint group, a technical handoff was prepared following management direction on **July 8, 2026** to ensure project continuity. Project documentation continued to be maintained by **Jephete Fenelon**, while **Joshua Ugborugbo** performed the remaining endpoint validation, Cloudflare verification, and DNS investigation

activities under management guidance. This collaborative approach ensured that implementation progress continued while maintaining a complete and accurate technical record of the project.

Current Project Status

Project Phase: Open – Operational Support, Final DNS Configuration, and Ongoing Production Validation

Successfully Completed

- Project planning and solution design
- Endpoint preparation and system configuration
- ContaCam deployment and configuration
- HP TrueVision HD Camera implementation
- Browser-based local monitoring
- Authentication and security implementation
- Internal web server configuration and validation
- Windows power optimization
- Cloudflare Zero Trust deployment
- Cloudflare Tunnel deployment
- Cloudflare connector (**cloudflared**) installation
- Cloudflare Tunnel validation and health verification
- Published Application configuration
- Local endpoint validation
- DNS investigation and root cause analysis
- Comprehensive testing, troubleshooting, and technical documentation
- Technical handoff and collaborative project transition

Remaining Activities

- Complete authoritative DNS configuration within the active DNS provider.
 - Validate external hostname resolution for **cameras.twignaturals.com**.
 - Perform final external camera accessibility testing.
 - Continue production validation and operational monitoring.
 - Document future implementation updates as additional work is completed.
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Project Outcome

As of **July 17, 2026**, the implementation, configuration, testing, troubleshooting, research, and documentation completed during my internship have established a secure and reliable foundation for the **TN-Laptop1 Security Camera Deployment and Remote Monitoring Project**.

Following the documented technical handoff, the remaining production validation and DNS-related activities continue under the direction of the assigned project team. The outstanding work is primarily related to production DNS configuration and final external accessibility verification rather than deficiencies within the deployed monitoring solution.

This document represents the technical implementation guide and project record prepared during my internship. At management's direction, it will remain an active implementation guide to support knowledge transfer, operational continuity, future maintenance, troubleshooting, infrastructure updates, and ongoing implementation activities performed by the Twigs Naturals IT team.

5. Project Environment

The **TN-Laptop1 Security Camera Deployment and Remote Monitoring Project** was implemented within the **Twigs Naturals** production environment to establish a secure, reliable, and continuously operating camera monitoring solution. The implementation utilized the existing workstation infrastructure together with the integrated **HP TrueVision HD Camera**, **ContaCam**, **Cloudflare Zero Trust**, and **Cloudflare Tunnel** technologies to support both local monitoring and the foundation for secure remote access.

The project environment was designed to support continuous browser-based local monitoring while evaluating secure cloud-based remote access technologies. Throughout the implementation, the workstation was managed through the **Action1 Endpoint Management** platform to facilitate remote administration, troubleshooting, validation, and operational maintenance.

As the project progressed, collaboration among team members supported the continuation of the remaining implementation activities. Following management direction, project documentation continued to be maintained by **Jephete Fenelon**, while **Joshua Ugborugbo** completed additional endpoint validation, Cloudflare verification, and DNS investigation activities. This implementation guide remains an active technical document to support ongoing operational activities, future enhancements, troubleshooting, and knowledge transfer.

6. Environment Information

Item	Details
Project Name	TN-Laptop1 Security Camera Deployment and Remote Monitoring
Organization	Twigs Naturals
Device Name	TN-Laptop1
Device Purpose	Production Workstation with Integrated Security Camera
Responsible Party	Nigel Jordan
Primary Documentation	Jephete Fenelon
Remote Validation Support	Joshua Ugborugbo
Operating System	Windows 11
Camera Software	ContaCam Version 9.9.27
Camera Hardware	HP TrueVision HD Camera
Remote Management Platform	Action1 Endpoint Management
Cloud Platform	Cloudflare Zero Trust
Remote Access Technology	Cloudflare Tunnel
Deployment Location	Internal Production Environment
Monitoring Type	Browser-Based Local Monitoring with Future Secure Remote Access
Authentication	Username and Password Protected
Current Operational Status	Local Monitoring Fully Operational – Cloudflare Tunnel Operational – Project Remains Open for Ongoing DNS Configuration, Production Validation, and Operational Support

6. RISK ASSESSMENT

Risk Assessment

A comprehensive risk assessment was conducted before and throughout the implementation to identify operational, security, infrastructure, and administrative risks that could affect the deployment, validation, and long-term operation of the TN-Laptop1 Security Camera System. Appropriate mitigation strategies were implemented wherever possible, while remaining infrastructure dependencies were documented to support ongoing project management, operational continuity, and future production deployment.

Risk Assessment Summary

Risk	Potential Impact	Mitigation Strategy	Current Status
Unauthorized Camera Access	Unauthorized viewing of monitoring data	Username authentication and password protection implemented	Mitigated
Power Interruption	Loss of monitoring capability	Windows power settings optimized to prevent sleep mode	Mitigated
Software Failure	Loss of recording or monitoring services	Application validation, startup testing, and operational verification completed	Mitigated
Network Connectivity Loss	Temporary interruption of monitoring services	Local network validation and browser testing completed	Mitigated
External Access Limitations	Public access unavailable	Cloudflare Tunnel implementation, DNS investigation, and production validation	Pending DNS Configuration

Public IP Address Changes	Loss of external connectivity	Dynamic DNS and Cloudflare Tunnel architecture evaluated	Low Risk
DNS Configuration	Hostname not publicly resolvable	Root cause identified; authoritative DNS provider requires configuration	Pending
Cloudflare Nameserver Delegation	Cloudflare DNS zone inactive	DNS investigation completed and documented	Pending Administrative Action
Production Validation	Final verification before production use	Remaining validation transitioned following project handoff	In Progress

Overall Risk Assessment

The implementation has successfully mitigated the primary technical risks associated with deploying the TN-Laptop1 Security Camera System. Local monitoring functionality, browser-based monitoring, authentication controls, recording services, and Cloudflare Tunnel deployment have all been successfully implemented and validated.

Subsequent endpoint validation and DNS investigation confirmed that the deployed monitoring solution is operating correctly within the local environment. The remaining project risks are no longer associated with the endpoint, camera application, or Cloudflare Tunnel implementation, but instead relate to completion of the production DNS configuration and final external accessibility verification.

The overall project risk is therefore considered low, with the remaining work consisting primarily of administrative infrastructure configuration, ongoing production validation, and operational support rather than remediation of technical failures. The implementation guide remains active to document future updates and support continued project operations.

7. SYSTEM INFORMATION

System Information

The following information summarizes the production environment utilized throughout the planning, implementation, configuration, validation, troubleshooting, Cloudflare deployment, DNS investigation, and operational assessment of the **TN-Laptop1 Security Camera Deployment and Remote Monitoring Project**.

This configuration represents the established baseline environment and serves as the reference for future maintenance, troubleshooting, validation, operational support, and project continuity.

System Configuration Summary

Item	Details
Device Name	TN-Laptop1
Device Purpose	Production Workstation with Integrated Security Camera
Organization	Twigs Naturals
Operating System	Windows 11
Camera Software	ContaCam Version 9.9.27
Camera Hardware	HP TrueVision HD Camera
Remote Management Platform	Action1 Endpoint Management
Cloud Platform	Cloudflare Zero Trust
Remote Access Solution	Cloudflare Tunnel
Responsible Party	Nigel Jordan
Primary Implementation & Documentation	Jephete Fenelon
Remote Validation Support	Joshua Ugborugbo
Initial Deployment Date	June 15, 2026
Last Documentation Update	July 17, 2026
Current Project Status	Open – Local Monitoring Operational, Cloudflare Tunnel Operational, Ongoing DNS Configuration, Production Validation, and Operational Support

8. PRE-DEPLOYMENT ACTIVITIES

Pre-Deployment Activities

Prior to implementation, comprehensive planning, preparation, and validation activities were completed to ensure that the TN-Laptop1 workstation was suitable for deployment as a dedicated security monitoring platform. These activities established a stable implementation environment, reduced deployment risk, and provided the technical foundation for successful installation, configuration, operational validation, Cloudflare implementation, and long-term system reliability.

Pre-Deployment Activities Performed

- Verified endpoint connectivity through the Action1 Endpoint Management platform.
- Confirmed successful remote management functionality.
- Reviewed project objectives, implementation requirements, and deployment scope.
- Validated functionality of the integrated HP TrueVision HD Camera using the native Windows Camera application.
- Confirmed endpoint accessibility and required administrative privileges.
- Updated Action1 endpoint comments to accurately reflect device ownership and business purpose.
- Verified workstation availability for deployment activities.
- Confirmed operating system readiness and software compatibility.
- Verified local network connectivity to support browser-based monitoring and future secure remote access implementation.
- Prepared technical implementation documentation and project records prior to deployment activities.

Action1 Endpoint Documentation

Endpoint Comment

Production Workstation with Integrated Security Camera – Nigel Jordan

Pre-Deployment Outcome

All pre-deployment requirements were successfully completed before software installation. The workstation was confirmed to be operational, properly configured, and suitable for implementation. These preparation activities established a stable foundation for the successful

deployment, configuration, validation, troubleshooting, Cloudflare implementation, DNS investigation, and ongoing operational support completed throughout the project.

9. SOFTWARE INSTALLATION

9.1 Zoom Removal

Overview

Prior to deploying the monitoring solution, Zoom was removed from the workstation because it was no longer required for its operational purpose. Removing unnecessary software reduced resource utilization, minimized potential conflicts with camera resources, and simplified long-term system administration.

Removal Activities

- Reviewed installed applications.
- Confirmed Zoom was no longer required.
- Uninstalled Zoom through Windows application management.
- Verified successful removal.
- Confirmed no impact to operating system stability or camera functionality.

Installation Result

Item	Status
Application Removed	Zoom
Removal Status	Successful
Webcam Functionality	Verified
System Stability	Verified

Outcome

Zoom was successfully removed without affecting operating system stability or webcam functionality. Removing unnecessary software reduced resource utilization and prepared the workstation for deployment as a dedicated security monitoring endpoint.

9.2 ContaCam Installation

Overview

ContaCam Version **9.9.27** was selected as the primary monitoring platform because it provides browser-based monitoring, recording capabilities, authentication controls, motion detection, and an integrated web server suitable for both local monitoring and future secure remote access.

Installation Activities

- Downloaded the official ContaCam installation package.
- Installed the application successfully.
- Launched ContaCam.
- Verified application stability.
- Confirmed successful detection of the HP TrueVision HD Camera.
- Verified live camera functionality.
- Configured monitoring and recording settings.
- Enabled the integrated web server.
- Completed initial operational validation.

Installation Results

Validation Item	Result
Software Installation	Successful
Application Launch	Successful
Camera Detection	Successful
Live Video Feed	Successful
Recording Functionality	Successful
Integrated Web Server	Successful
Initial Operational Validation	Successful

Outcome

ContaCam was successfully deployed, configured, and validated. The application demonstrated stable performance throughout implementation and served as the foundation for the TN-Laptop1

Security Camera Deployment and Remote Monitoring Project. Subsequent validation confirmed that the application, integrated camera, recording services, browser-based monitoring, and local web server operated reliably throughout the implementation lifecycle.

10. SYSTEM CONFIGURATION

10.1 Power Configuration

Overview

Windows power management settings were optimized to support uninterrupted camera monitoring while reducing unnecessary energy consumption. The configuration was designed to ensure continuous operation of the monitoring system while allowing normal display power management, thereby maintaining recording services and browser-based monitoring availability during extended periods of operation.

Validation Results

- Camera operation remained active after display timeout.
- Recording services continued without interruption.
- Browser-based monitoring remained accessible throughout testing.
- No interruption to monitoring services was observed during extended operation.

Outcome

Windows power management was successfully optimized to support continuous monitoring while maintaining normal display power management. Operational testing confirmed that power settings did not interfere with camera operation, recording functionality, or browser-based monitoring services throughout the implementation lifecycle.

10.2 Camera Configuration

Overview

The integrated **HP TrueVision HD Camera** was configured within **ContaCam** to provide continuous monitoring, browser-based viewing, motion recording, and reliable long-term

operation. Configuration settings were optimized to support production monitoring while providing a stable foundation for secure remote access through Cloudflare technologies.

Configuration Activities

- Configured the integrated HP TrueVision HD Camera.
- Verified camera detection and image quality.
- Enabled automatic startup.
- Configured continuous monitoring and recording.
- Enabled timestamp overlay.
- Configured recording retention policy.
- Validated camera operation following application restart.

Validation Results

- Camera detected successfully.
- Live video displayed correctly.
- Recording services operated as expected.
- Monitoring services remained active after restart.
- Continuous monitoring successfully validated.

Outcome

Camera configuration was successfully completed and validated for continuous production use. Throughout subsequent implementation, troubleshooting, and validation activities, the camera remained stable and consistently supported browser-based monitoring, recording services, and local operational testing.

10.3 Network Configuration

Overview

The network configuration was established to support browser-based monitoring within the local network while preparing the environment for secure external access through Cloudflare Zero Trust and Cloudflare Tunnel technologies.

Outcome

All local networking requirements were successfully validated. Local browser-based monitoring remained fully operational throughout implementation. Subsequent DNS investigation confirmed that the remaining limitation affecting external accessibility was related to authoritative DNS configuration rather than local network configuration or endpoint operation.

10.4 Remote Monitoring Configuration

Outcome

The browser-based monitoring solution was successfully implemented and validated. Local monitoring remained fully operational throughout the project and provided the foundation for secure remote monitoring through Cloudflare Tunnel. Validation confirmed successful local access, secure tunnel connectivity, and Published Application configuration, with remaining production activities limited to DNS configuration and external accessibility verification.

10.5 Authentication and Security

Outcome

Authentication and security controls were successfully implemented and validated. Access to the monitoring environment is protected through authenticated user access, providing an appropriate level of security for operational monitoring while supporting secure browser-based access through the Cloudflare Zero Trust environment.

11. EXTERNAL ACCESS ASSESSMENT

Overview

The TN-Laptop1 Security Camera System was designed to support secure remote monitoring while preserving the organization's existing network, website, email, and DNS infrastructure. Although local monitoring and Cloudflare Tunnel deployment were successfully implemented and validated, secure public access required additional DNS validation and infrastructure assessment.

Throughout the project, multiple implementation approaches were evaluated to provide secure remote access without exposing the internal network directly to the Internet.

Identified Risks

Risk	Potential Impact
Router firewall restrictions	Prevent external access to monitoring services
Network Address Translation (NAT) limitations	Block inbound connections
ISP routing restrictions (CGNAT)	Prevent direct external connectivity
Dynamic public IP address changes	Interrupt external access
DNS configuration	Prevent hostname resolution
Endpoint validation requirements	Delay final implementation verification

Mitigation Strategies

The following mitigation strategies were implemented or evaluated throughout the project:

- Reviewed router firewall configuration.
 - Validated DHCP reservation and local network configuration.
 - Reviewed port forwarding requirements.
 - Investigated WAN connectivity and public IP configuration.
 - Researched Dynamic DNS (DDNS) alternatives.
 - Evaluated static public IP recommendations.
 - Implemented Cloudflare Zero Trust.
 - Deployed Cloudflare Tunnel.
 - Researched secure DNS implementation through Cloudflare Tunnel.
 - Evaluated VPN-based remote access as a future enhancement.
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Current Status

The following implementation activities have been successfully completed:

- Local monitoring fully operational.
- Browser-based monitoring validated.
- Cloudflare Zero Trust deployed.
- Cloudflare Tunnel successfully deployed and validated.
- Published Application configured.
- Secure tunnel communication established.

- Endpoint validation completed.
- DNS investigation and root cause analysis completed.

Remaining activities include:

- Complete authoritative DNS configuration within the active DNS provider.
 - Validate external hostname resolution for **cameras.twignaturals.com**.
 - Perform production accessibility verification.
 - Continue operational monitoring and implementation updates as additional work is completed.
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12. SYSTEM TESTING AND VALIDATION

Overview

Comprehensive testing and validation activities were performed throughout the implementation lifecycle to verify system functionality, operational stability, security, Cloudflare connectivity, DNS configuration, and readiness for production deployment. Validation activities included local endpoint verification, browser-based monitoring, Cloudflare Tunnel implementation, DNS investigation, root cause analysis, and external connectivity assessment to ensure the deployed monitoring solution operated as designed.

Validation Summary

Validation Test	Result
Camera Detection	Passed
Live Video Feed	Passed
Recording Functionality	Passed
Browser-Based Monitoring	Passed
Authentication Verification	Passed
Internal Web Server	Passed
Local Monitoring	Passed

HTTP Port Validation	Passed
HTTPS Port Validation	Passed
System Restart Validation	Passed
Continuous Operation	Passed
Cloudflare Tunnel Deployment	Passed
Cloudflare Tunnel Service Validation	Passed
Published Application Configuration	Passed
Tunnel Health Verification	Passed
Local Endpoint Validation	Passed
DNS Investigation	Completed
Root Cause Analysis	Completed

Additional Validation Activities

The following validation activities were completed throughout the implementation lifecycle:

- Verified automatic application startup following system reboot.
- Confirmed timestamp functionality.
- Validated recording retention policy.
- Verified continuous monitoring following display timeout.
- Tested browser-based monitoring functionality.
- Confirmed application stability.
- Verified authentication controls.
- Confirmed monitoring accessibility through the local web interface.
- Validated Cloudflare Tunnel connectivity and service health.
- Confirmed Published Application routing.
- Performed local and external connectivity testing.
- Validated Cloudflare Tunnel service operation.
- Investigated authoritative DNS configuration.
- Confirmed DNS root cause through infrastructure validation.

- Verified that the remaining external accessibility limitation was related to authoritative DNS configuration rather than endpoint operation.

Overall Validation Outcome

All planned implementation objectives relating to the local monitoring solution were successfully completed. Camera functionality, recording services, browser-based monitoring, authentication controls, application stability, Cloudflare Zero Trust deployment, Cloudflare Tunnel implementation, and local endpoint validation were successfully implemented and verified.

Additional endpoint validation, Cloudflare verification, and DNS investigation confirmed that the deployed monitoring solution is operating correctly within the local environment. The remaining limitation preventing public access is associated with authoritative DNS configuration and nameserver delegation rather than the deployed endpoint, monitoring application, or Cloudflare Tunnel implementation.

The implementation, validation, troubleshooting, and documentation activities completed during my internship have been successfully documented. The remaining work is limited to production DNS configuration, ongoing external accessibility verification, operational support, and future implementation updates. In accordance with management direction, this implementation guide remains an active technical document to support continued project development and knowledge transfer.

13. EXTERNAL ACCESS INVESTIGATION

Investigation Overview

Following the successful deployment of the local monitoring environment, a comprehensive investigation was conducted to determine why external access to the TN-Laptop1 Security Camera System remained unavailable.

The objective of the investigation was to determine whether the limitation originated from the workstation, ContaCam application, local network, Cloudflare implementation, DNS configuration, or external network infrastructure.

Investigation Activities

The following investigative activities were completed throughout the project:

- Verified normal operation of the HP TrueVision HD Camera.
- Confirmed live video streaming.
- Validated recording functionality.
- Verified ContaCam Internal Web Server operation.
- Confirmed HTTP and HTTPS configuration.
- Tested browser access using localhost and 127.0.0.1.
- Verified user authentication.
- Validated browser-based monitoring.
- Verified local monitoring services.
- Confirmed workstation IPv4 configuration.
- Verified default gateway configuration.
- Reviewed router configuration.
- Reviewed DHCP configuration.
- Investigated port forwarding requirements.
- Reviewed Windows Firewall configuration.
- Validated Cloudflare Tunnel deployment.
- Verified Published Application routing.
- Investigated DNS implementation requirements.
- Performed authoritative DNS validation.
- Performed external connectivity testing.
- Conducted DNS infrastructure investigation and root cause analysis.

Investigation Findings

The investigation confirmed that:

- The workstation operated normally.
- ContaCam functioned correctly.
- Recording services operated normally.
- Authentication controls functioned correctly.
- Browser-based monitoring operated successfully.
- Local monitoring remained fully operational.
- Cloudflare Tunnel deployment was successful.
- Cloudflare Tunnel service was operational.
- Published Application routing functioned correctly.
- The Cloudflare Tunnel implementation was not the cause of the external accessibility limitation.
- The authoritative DNS provider remained Squarespace/NS1.
- The Cloudflare DNS zone was inactive due to invalid nameserver delegation.

No evidence indicated a failure of the deployed monitoring solution.

Technical Assessment

Technical investigation confirmed that the remaining implementation activities are associated with authoritative DNS configuration and final production validation rather than workstation configuration, application functionality, or Cloudflare Tunnel deployment.

Earlier testing considered possible upstream network limitations, including Carrier-Grade NAT (CGNAT). However, subsequent DNS validation identified the authoritative DNS configuration as the primary infrastructure dependency preventing public hostname resolution.

Outcome

Local monitoring functionality has been fully implemented and validated.

Cloudflare Zero Trust and Cloudflare Tunnel have been successfully deployed and verified.

DNS investigation and root cause analysis have been completed.

The remaining implementation activities are limited to updating the authoritative DNS configuration, validating public hostname resolution, and completing final production verification.

14. TECHNICAL TROUBLESHOOTING ACTIVITIES

Overview

Comprehensive troubleshooting activities were performed throughout the implementation lifecycle to verify operational stability, identify root causes, resolve implementation issues, and support the successful deployment of the TN-Laptop1 Security Camera Deployment and Remote Monitoring Project.

Troubleshooting Activities

The following troubleshooting activities were completed:

- Restored interrupted Action1 remote sessions.
- Re-established endpoint connectivity.
- Validated camera operation following system reboot.
- Investigated ContaCam startup behavior.
- Verified browser accessibility.
- Reviewed Windows Firewall configuration.
- Verified required firewall rules.

- Investigated local and external connectivity.
 - Configured authentication controls.
 - Verified Internal Web Server functionality.
 - Confirmed live video streaming.
 - Reviewed local network configuration.
 - Identified router management interface.
 - Reviewed router configuration.
 - Investigated WAN connectivity.
 - Implemented Cloudflare Zero Trust.
 - Deployed Cloudflare Tunnel.
 - Installed and validated the cloudflared connector.
 - Validated Cloudflare Tunnel service health.
 - Verified Published Application configuration.
 - Performed DNS research and infrastructure validation.
 - Investigated authoritative DNS configuration.
 - Identified the DNS root cause affecting external hostname resolution.
 - Prepared and completed a technical handoff to support project continuity and collaborative implementation.
-

Outcome

Troubleshooting activities confirmed that the deployed monitoring solution is operating correctly within the local environment. Endpoint validation, Cloudflare verification, and DNS investigation confirmed that the remaining production dependency is related to authoritative DNS configuration rather than deficiencies within the deployed monitoring solution.

The troubleshooting phase successfully identified the root cause of the external accessibility limitation and established a clear path for completing production deployment following the required DNS configuration changes. The project is therefore considered technically complete, with the remaining activities limited to infrastructure administration and final production verification.

15. PROJECT OUTCOMES AND KNOWLEDGE GAINED

Project Outcomes

This project provided practical experience in:

- Endpoint management
 - Security camera deployment
 - Browser-based monitoring
 - Cloudflare Zero Trust
 - Cloudflare Tunnel implementation
 - DNS implementation planning
 - Authentication and access control
 - Windows Firewall configuration
 - Router and network troubleshooting
 - Remote endpoint administration
 - Technical documentation
 - Root cause analysis
 - Operational validation
 - Project coordination and technical handoff
-

Key Lessons Learned

The project reinforced several important operational practices:

- Validate local functionality before investigating external connectivity.
 - Comprehensive documentation improves project continuity and troubleshooting efficiency.
 - Authentication controls are essential for protecting monitoring systems.
 - Cloudflare Tunnel provides a secure alternative to traditional port forwarding.
 - Infrastructure dependencies can significantly affect remote access implementation.
 - Collaboration between technical team members supports successful project completion.
 - Clear technical handoffs improve continuity when responsibilities transition between team members.
-

16. CURRENT PROJECT STATUS

Project Status (Updated July 8, 2026)

Successfully Completed

- ContaCam deployment and configuration.

- Camera implementation.
- Browser-based monitoring.
- Authentication implementation.
- Internal Web Server configuration.
- Continuous recording configuration.
- Windows power optimization.
- Router review.
- DHCP validation.
- Windows Firewall validation.
- Cloudflare Zero Trust deployment.
- Cloudflare Tunnel deployment.
- Published Application configuration.
- Technical documentation.
- Comprehensive testing and troubleshooting.

Current Activities

Following management direction, the remaining endpoint validation activities are being completed collaboratively. Documentation continues to be maintained by Jephete Fenelon, while the remaining remote validation activities will be performed through Action1 Remote Connect by Joshua Ugborugbo.

Remaining Activities

- Verify the local ContaCam service.
- Confirm service address and listening port.
- Complete endpoint validation.
- Validate external DNS configuration.
- Perform final external camera accessibility testing.
- Update documentation with final validation results.
- Complete project closure.

Current Project Status

Research and Validation Phase – Final Validation in Progress

17. PHASE III – CLOUDFLARE ZERO TRUST IMPLEMENTATION

Overview

Phase III focused on implementing **Cloudflare Zero Trust** and **Cloudflare Tunnel** technologies to establish a secure remote access solution for the TN-Laptop1 Security Camera System. The objective was to evaluate a cloud-based approach that would provide secure external access without exposing the organization's internal network directly to the Internet.

Project Objectives

- Implement Cloudflare Zero Trust.
 - Deploy Cloudflare Tunnel.
 - Install and configure the **cloudflared** connector.
 - Validate secure tunnel connectivity.
 - Configure a Published Application.
 - Research DNS integration requirements.
 - Evaluate secure remote access without traditional port forwarding.
 - Document implementation findings and recommendations.
-

18. CLOUDFLARE TUNNEL IMPLEMENTATION

Overview

Cloudflare Tunnel was successfully implemented as the preferred secure remote access solution for the TN-Laptop1 Security Camera System. The implementation established secure outbound communication between the local monitoring service and the Cloudflare Zero Trust environment while maintaining the organization's existing website, email, and DNS infrastructure.

Implementation Activities

The following activities were successfully completed:

- Validated the existing Cloudflare Tunnel configuration.
- Successfully deployed the Cloudflare Tunnel.
- Installed and configured the **cloudflared** connector.
- Verified secure communication with Cloudflare Zero Trust.
- Confirmed tunnel health and active replica status.
- Configured the Published Application for **cameras.twignaturals.com**.

- Verified hostname mapping to the local ContaCam service.
 - Validated secure routing between Cloudflare and the monitoring application.
 - Performed external validation testing.
 - Researched DNS implementation within the existing Squarespace-managed environment.
-

Cloudflare Deployment Status

Component	Status
Cloudflare Zero Trust	Completed
Cloudflare Tunnel	Completed
cloudflared Installation	Completed
Tunnel Connectivity	Completed
Tunnel Health	Completed
Published Application	Completed
ContaCam Service Mapping	Completed
DNS Validation	Pending
External Camera Validation	Pending

Technical Assessment

Testing confirmed that the Cloudflare implementation is functioning correctly. Secure tunnel communication has been established, and the published application has been successfully configured.

The remaining implementation work is limited to final endpoint verification, DNS validation, and external accessibility testing.

19. OPERATIONAL STATUS UPDATES

July 5, 2026 – Action1 Remote Connect Dependency Identified

Additional troubleshooting determined that final validation of the local ContaCam service required remote access through the Action1 Remote Connect feature.

Because the assigned Action1 account did not have Remote Connect permissions for the TN endpoint group, the remaining endpoint validation activities could not be completed.

No Cloudflare deployment issues were identified during this investigation.

July 7, 2026 – Follow-up Validation

A follow-up validation attempt confirmed that Action1 Remote Connect permissions remained unavailable.

The following activities therefore remained outstanding:

- Verification of the local ContaCam service.
- Confirmation of the configured listening ports.
- Browser accessibility validation.
- Final Cloudflare Tunnel verification.
- External camera accessibility testing.

No additional Cloudflare configuration changes were performed because local endpoint validation could not be completed.

July 8, 2026 – Project Transition and Technical Handoff

Following management direction, responsibility for the remaining endpoint validation activities transitioned to a collaborative approach.

Project documentation continues to be maintained by **Jephete Fenelon**, while **Joshua Ugborugbo** will perform the remaining endpoint validation activities using **Action1 Remote Connect**.

A technical handoff was prepared summarizing:

- Completed implementation activities.
- Current project status.

- Remaining validation requirements.
- Recommended verification procedures.
- Outstanding implementation dependencies.

The handoff was prepared to support efficient project completion while maintaining continuity of technical documentation.

July 9, 2026 – Technical Validation and Endpoint Verification

Overview

Following the technical handoff completed on July 8, 2026, Joshua Ugborugbo continued the remaining endpoint validation activities using Action1 Remote Connect. The objective of this phase was to verify the operational status of the deployed monitoring solution, confirm the Cloudflare Tunnel implementation, validate the local camera services, and determine whether any remaining issues existed within the endpoint configuration.

This validation represented the continuation of the implementation project rather than a new deployment. All activities were performed against the existing TN-Laptop1 Security Camera environment that had previously been deployed and documented. The purpose was to verify operational readiness and identify any remaining infrastructure dependencies preventing successful external access.

Validation Activities Performed

Joshua completed the following validation activities:

- Reviewed the existing ContaCam deployment.
- Verified successful operation of the HP TrueVision HD Camera.
- Confirmed the live camera feed was functioning correctly.
- Reviewed the ContaCam Global Settings configuration.
- Verified camera recording configuration.
- Validated advanced recording settings.
- Confirmed HTTP (Port 8800) configuration.
- Confirmed HTTPS (Port 8443) configuration.
- Tested browser access using localhost.
- Tested browser access using 127.0.0.1.
- Verified successful authentication through the ContaCam web interface.

- Confirmed operation of the embedded web server.
 - Started and validated the Cloudflare Tunnel service.
 - Verified tunnel health within Cloudflare Zero Trust.
 - Confirmed the Published Application remained correctly configured.
 - Reviewed Cloudflare DNS configuration.
 - Investigated external DNS resolution.
 - Documented remaining infrastructure dependencies affecting external connectivity.
-

Validation Results

The endpoint validation confirmed that all locally deployed components were operating as expected.

Successful validation included:

- ContaCam application
- Integrated HP TrueVision HD Camera
- Continuous live video monitoring
- Internal web server
- HTTP service
- HTTPS service
- User authentication
- Browser-based monitoring
- Cloudflare Tunnel service
- Published Application configuration
- Tunnel connectivity

Testing confirmed that the deployed monitoring solution was functioning correctly within the local environment.

Cloudflare Validation

During validation, the Cloudflare Tunnel service was initially identified in a stopped state.

Using administrative privileges, Joshua started the **cloudflared** service and verified that it successfully entered the **RUNNING** state.

Additional validation confirmed:

- Tunnel Status: Healthy

- Active Replica: 1
- Published Hostname: cameras.twignaturals.com
- Tunnel Connectivity: Operational
- Published Application: Verified

These results confirmed that the secure tunnel deployment remained operational and capable of supporting external connectivity once DNS activation is completed.

DNS Assessment

The Cloudflare configuration was reviewed to determine why public access remained unavailable.

Validation confirmed that the DNS record for **cameras.twignaturals.com** had already been created and correctly associated with the Cloudflare Tunnel.

However, the parent domain **twignaturals.com** remained in a pending state because the authoritative nameservers had not yet been updated at the domain registrar.

Cloudflare therefore displayed a pending status while awaiting completion of nameserver propagation.

Because Cloudflare was not yet the authoritative DNS provider, public DNS resolution for the hostname remained unavailable, resulting in an NXDOMAIN response during external testing.

This condition was determined to be an infrastructure dependency rather than an issue with the deployed endpoint or Cloudflare Tunnel configuration.

Technical Assessment

The July 9 validation confirmed that the deployment activities completed during the implementation phase had been successfully configured and remained operational.

No issues were identified with:

- ContaCam
- Camera hardware
- Local monitoring
- Authentication
- Internal web server
- Cloudflare Tunnel

- Published Application
- Secure tunnel connectivity

The remaining work required completion of the domain registrar nameserver update before external DNS resolution and public access could be validated.

Project Status Update – July 9, 2026

The project remained in the final validation phase following completion of the endpoint verification activities.

The implementation team confirmed that all endpoint-side configuration tasks had been completed successfully. The only outstanding dependency preventing production operation was the completion of the authoritative nameserver update at the domain registrar.

Following DNS propagation, external access to **cameras.twignaturals.com** can be validated and the project can proceed toward production closure.

July 13, 2026 – DNS Infrastructure Validation and Root Cause Analysis

Overview

Following the successful endpoint validation completed on July 9, 2026, additional DNS investigation was performed by **Joshua Ugborugbo** on **July 13, 2026** to determine why the hostname **cameras.twignaturals.com** remained inaccessible from the public Internet despite the successful deployment of the Cloudflare Tunnel and Published Application.

The objective of this phase was to identify the authoritative DNS provider for the **twignaturals.com** domain, verify the active DNS infrastructure, evaluate the Cloudflare DNS configuration, and determine the root cause of the unresolved hostname.

Objective

The DNS validation was performed to:

- Verify the authoritative DNS provider for **twignaturals.com**.
 - Determine why **cameras.twignaturals.com** returned an **NXDOMAIN** response.
 - Confirm whether the Cloudflare DNS zone was actively serving the domain.
 - Validate synchronization between the active DNS infrastructure and Cloudflare.
 - Identify the remaining infrastructure dependency preventing external access to the security camera.
-

DNS Validation Activities

Joshua completed the following technical validation activities:

- Performed public DNS resolution testing using **nslookup**.
 - Queried authoritative NS1 name servers.
 - Queried Squarespace authoritative name servers.
 - Compared DNS responses from both providers.
 - Reviewed the Cloudflare DNS configuration.
 - Reviewed Cloudflare domain status.
 - Validated nameserver delegation status.
 - Investigated the cause of the **NXDOMAIN** response.
 - Documented the DNS infrastructure findings.
-

Evidence 1 – NS1 Authoritative DNS Validation

Direct DNS queries were performed against an NS1 authoritative nameserver using:

```
nslookup twignaturals.com dns1.p02.nsone.net
```

The query successfully returned the public IP address assigned to **twignaturals.com**, confirming that the NS1 infrastructure was actively serving authoritative DNS records for the production domain.

The returned address was:

67.208.40.105

This result confirmed that the production DNS zone remained active within the NS1 infrastructure.

Evidence 2 – Squarespace DNS Validation

Additional validation was performed by querying a Squarespace authoritative nameserver using:

nslookup twigsnaturals.com ns01.squarespacedns.com

The query returned the same public IP address:

67.208.40.105

The identical response confirmed that Squarespace and NS1 were synchronized and serving the same authoritative DNS zone.

This validation demonstrated that the production DNS environment continued to operate through Squarespace's managed DNS infrastructure.

Evidence 3 – Cloudflare DNS Review

The Cloudflare dashboard was reviewed to determine whether the DNS zone configured within Cloudflare was actively serving the domain.

The review confirmed that a Cloudflare DNS zone for **twigsnaturals.com** had previously been created and contained Cloudflare-assigned nameservers.

Although the zone appeared fully configured, additional validation was required to determine whether it was actually delegated by the domain registrar.

Evidence 4 – Cloudflare Nameserver Validation

Further investigation of the Cloudflare domain status identified the following condition:

Status: Invalid Nameservers

This status confirmed that the Cloudflare-assigned nameservers had **not** been delegated by the domain registrar.

As a result:

- Cloudflare was not acting as the authoritative DNS provider.
- DNS records created within the Cloudflare zone were not publicly available.
- Public DNS requests continued to be answered by Squarespace/NS1.

This finding eliminated Cloudflare Tunnel configuration as the source of the problem.

Root Cause Analysis

The DNS investigation determined that the Cloudflare Tunnel deployment and Published Application configuration were functioning correctly.

The inability to resolve **cameras.twignaturals.com** was caused by DNS delegation rather than endpoint configuration.

Because the authoritative nameservers remained assigned to the Squarespace-managed DNS infrastructure, any DNS records configured within the inactive Cloudflare DNS zone were ignored by public DNS resolvers.

Consequently, requests for **cameras.twignaturals.com** returned an **NXDOMAIN** response because the required hostname did not exist within the active authoritative DNS zone.

Technical Assessment

The July 13 validation confirmed that the remaining issue preventing external camera access was not related to:

- ContaCam configuration
- Camera hardware
- Local monitoring services
- Authentication
- Cloudflare Tunnel deployment
- Published Application configuration
- Endpoint configuration
- Windows networking

Instead, the remaining dependency was identified as the active DNS infrastructure managing **twignaturals.com**.

The production domain continued to use Squarespace/NS1 as its authoritative DNS provider, while the Cloudflare DNS zone remained inactive due to invalid nameserver delegation.

Project Status Update – July 13, 2026

The DNS investigation successfully identified the root cause preventing public access to **cameras.twignaturals.com**.

Endpoint validation remained complete, and the Cloudflare Tunnel continued operating normally.

The project entered the next phase of implementation, requiring DNS configuration changes within the active authoritative DNS provider or delegation of the domain to Cloudflare before external access could be successfully completed.

July 16, 2026 – Project Continuation and Transition Update

Overview

Following the technical handoff completed on July 8, 2026, the remaining Cloudflare implementation and validation activities continued under the direction of Nigel Jordan and Joshua Ugborugbo. Additional work focused on completing the remaining Cloudflare Tunnel implementation, DNS validation, and secure remote access configuration required to support the TN-Laptop1 Security Camera Deployment and Remote Monitoring Project.

During this phase, Joshua continued working with management to complete the remaining implementation activities while following the recommended Cloudflare deployment procedures. Nigel provided additional implementation guidance, including a simplified Windows 10 Cloudflare Tunnel deployment methodology, to support the successful completion of the remaining project tasks.

These activities represent the continuation of the implementation beyond the initial deployment and validation phases documented throughout this guide.

Current Project Status

As of **July 16, 2026**, the project has successfully achieved the following milestones:

- ContaCam deployment and configuration completed.
- HP TrueVision HD Camera successfully configured and validated.
- Browser-based local monitoring fully operational.
- Authentication and security controls implemented.

- Internal web server configuration validated.
- Cloudflare Zero Trust environment configured.
- Cloudflare Tunnel successfully deployed.
- Published Application configured.
- Comprehensive testing, troubleshooting, and technical documentation completed.
- DNS investigation and root cause analysis completed.
- Technical handoff completed to support project continuity.

The remaining implementation activities continue to focus on final Cloudflare deployment tasks, DNS configuration, and production verification before full external access can be completed.

Project Transition

As my internship concludes on **July 17, 2026**, responsibility for the remaining implementation activities will continue with the assigned project team. Throughout the internship, I was responsible for the planning, implementation, configuration, testing, troubleshooting, documentation, and technical research supporting the TN-Laptop1 Security Camera Deployment and Remote Monitoring Project.

Following the documented technical handoff, Joshua Ugborugbo has continued the remaining implementation and validation activities under management direction to support the successful completion of the project.

This documentation has been maintained throughout the implementation lifecycle to provide a comprehensive technical record of the project, support knowledge transfer, and ensure continuity for future maintenance and project completion.

20. CURRENT PROJECT STATUS

Current Status

Project Phase: Open – Ongoing Implementation and Operational Support

Status Summary

The initial implementation, Cloudflare Tunnel deployment, system configuration, documentation, testing, and technical handoff completed during my internship have been documented in this guide. At management's direction, this implementation guide remains open to support future enhancements, troubleshooting, DNS updates, validation activities, and ongoing operational

support. Future implementation activities may be documented by Joshua Ugborugbo or other authorized members of the IT team as the project continues.

Successfully Completed

The following implementation activities have been successfully completed:

- Project planning and technical design
 - Endpoint preparation and system configuration
 - ContaCam deployment and configuration
 - HP TrueVision HD Camera implementation
 - Browser-based local monitoring
 - Authentication and security implementation
 - Internal Web Server configuration and validation
 - Windows power optimization
 - Recording functionality configuration and validation
 - Cloudflare Zero Trust deployment
 - Cloudflare Tunnel deployment
 - Cloudflare Tunnel (cloudflared) installation and validation
 - Cloudflare Tunnel health verification
 - Published Application configuration
 - Local endpoint validation
 - Cloudflare dashboard validation
 - DNS investigation and root cause analysis
 - Technical documentation
 - Comprehensive testing and troubleshooting
 - Technical handoff and project transition
-

Remaining Activities

The remaining project activities are administrative and infrastructure-related rather than technical implementation tasks.

The following activities remain before production deployment can be completed:

- Configure the required DNS record within the authoritative DNS provider (Squarespace/NS1) or delegate the domain to Cloudflare.

- Validate public hostname resolution for cameras.twigsnaturals.com.
- Perform final external camera accessibility verification.
- Complete production deployment validation.
- Continue operational support and document future implementation updates as additional work is completed.

The remaining project activities include production DNS configuration, final infrastructure validation, operational support, and completion of the remaining implementation and production verification activities. These activities will continue under the direction of the Twigs Naturals IT team as the implementation guide remains an active project document.

Remaining Validation Activities

- Verify the local ContaCam service through Action1 Remote Connect.
 - Confirm configured service address and listening ports.
 - Validate browser accessibility from the endpoint.
 - Validate external DNS configuration.
 - Perform external camera accessibility testing.
 - Update documentation with final validation findings.
 - Continue operational validation and update this implementation guide as additional implementation activities are completed.
-

Current Project Assessment

The TN-Laptop1 Security Camera Deployment and Remote Monitoring Project has successfully achieved its planned implementation objectives.

Comprehensive implementation, configuration, testing, troubleshooting, endpoint validation, Cloudflare verification, and DNS investigation confirmed that the deployed monitoring solution is operating correctly within the local environment.

No technical deficiencies were identified within the deployed monitoring solution. The remaining work is limited to completing the required authoritative DNS configuration and performing final production verification before secure external access can be enabled.

Following the documented technical handoff, project continuity has been maintained through collaborative implementation activities. This document serves as the complete technical implementation guide and project record developed throughout my internship and is intended to support future maintenance, operational continuity, knowledge transfer, and successful project completion.

21. APPENDIX – SUPPORTING EVIDENCE

The following supporting evidence should be retained with this document to provide validation of the implementation, testing, troubleshooting, and research activities performed throughout the project.

Action1 Documentation

- Action1 Endpoint Overview
- Endpoint Comment Configuration
- Remote Management Activities
- Endpoint Status Validation

Deployment Evidence

- Zoom Removal
- ContaCam Installation
- Camera Configuration
- Authentication Configuration
- Internal Web Server Configuration
- Browser-Based Monitoring
- Local Monitoring Validation
- Windows Power Configuration

Network Validation

- IPv4 Address Verification
- Windows Firewall Configuration
- Router Administration
- DHCP Configuration
- Port Forwarding Configuration
- WAN Address Validation

Cloudflare Implementation

- Cloudflare Zero Trust Configuration
- Cloudflare Tunnel Deployment
- cloudflared Installation
- Tunnel Health Verification
- Published Application Configuration
- DNS Research
- Cloudflare Dashboard
- Tunnel Validation Results

Document Maintenance

This implementation guide is maintained as a living technical document. While the implementation activities completed during my internship are documented herein, the guide is intended to remain open for future updates, operational improvements, troubleshooting activities, infrastructure changes, and technical validation.

At the direction of management, this document remains active to support ongoing operational support, knowledge transfer, and future implementation activities. Authorized members of the Twigs Naturals IT team may continue updating this document as additional work is performed.

This wording also ties directly to what Nigel told you: **leave it open**. It makes it clear to Joshua or anyone else on the IT team that they can continue updating the guide while preserving the work you completed during your internship.

22. CONCLUSION

Final Project Summary

The **TN-Laptop1 Security Camera Deployment and Remote Monitoring Project** successfully established a secure and reliable local monitoring solution utilizing **ContaCam**, the integrated **HP TrueVision HD Camera**, **Cloudflare Zero Trust**, and **Cloudflare Tunnel** technologies. The project was undertaken to design, implement, validate, and document a browser-based monitoring solution capable of supporting continuous local monitoring while providing the foundation for secure remote access.

Throughout the implementation lifecycle, comprehensive planning, deployment, system configuration, operational validation, troubleshooting, network analysis, Cloudflare implementation, DNS investigation, technical research, and documentation activities were successfully completed. Local monitoring capabilities, browser-based access, authentication controls, recording functionality, Windows power optimization, and secure Cloudflare Tunnel connectivity were implemented and validated, providing a stable and secure monitoring environment.

The Cloudflare Tunnel implementation established a secure foundation for future external monitoring while preserving the organization's existing website, email, and DNS infrastructure. Additional technical investigations confirmed that the deployed monitoring solution was operating correctly within the local environment. Subsequent DNS validation and root cause

analysis determined that the remaining external accessibility limitation was related to authoritative DNS configuration rather than the endpoint, Cloudflare Tunnel deployment, or application configuration.

Following management direction on **July 8, 2026**, responsibility for the remaining implementation and validation activities transitioned to a collaborative approach. Project documentation continued to be maintained by **Jephete Fenelon**, while **Joshua Ugborugbo** continued the remaining endpoint validation, Cloudflare implementation, and DNS verification activities under management guidance to support successful project completion. Additional implementation guidance provided by **Nigel Jordan** supported the continuation of the remaining Cloudflare deployment tasks and final production validation.

This document represents the technical implementation guide and project record prepared during my internship. At management's direction, the guide remains an active technical document to support ongoing implementation activities, knowledge transfer, operational continuity, troubleshooting, and future maintenance by the Twigs Naturals IT team.

Final Project Status

Project Phase: Open – Operational Support, Final DNS Configuration, and Ongoing Production Validation

Successfully Completed

- Project planning and technical design
- ContaCam deployment and configuration
- HP TrueVision HD Camera implementation
- Browser-based local monitoring
- Authentication and security implementation
- Internal web server configuration and validation
- Windows power optimization
- Camera recording configuration and validation
- Cloudflare Zero Trust deployment
- Cloudflare Tunnel deployment
- Cloudflare Tunnel service validation
- Published Application configuration
- Local endpoint validation
- DNS investigation and root cause analysis
- Comprehensive testing and troubleshooting
- Technical documentation and implementation guide
- Technical handoff and project transition

Remaining Activities

- Complete final DNS configuration within the authoritative DNS provider
 - Validate external hostname resolution for **cameras.twignaturals.com**
 - Perform final external camera accessibility testing
 - Complete production verification
 - Continue operational monitoring and document future implementation activities as required.
-

Revised Closing Statement

Closing Statement

As my internship concludes on **July 17, 2026**, I am grateful for the opportunity to contribute to the TN-Laptop1 Security Camera Deployment and Remote Monitoring Project and to be part of the Twigs Naturals IT team. This project provided valuable hands-on experience in endpoint management, cybersecurity, systems administration, Cloudflare Zero Trust, Cloudflare Tunnel implementation, DNS troubleshooting, technical documentation, and collaborative project delivery.

Although my internship has concluded, this implementation guide will remain an active technical document to support ongoing operational activities, knowledge transfer, future maintenance, and continued implementation efforts by the Twigs Naturals IT team.

I sincerely appreciate the guidance, mentorship, and opportunities provided throughout my internship. The knowledge and practical experience gained during this project have significantly strengthened my technical and professional skills and will serve as a strong foundation for my continued career in Information Technology and Cybersecurity.